



Task - 02

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Submitted to:

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Abstract

Crack detection is a key task in structural inspection and maintenance systems. In this work, three convolutional neural network models (CNN)—VGG, ResNet50 and EfficientNet—are evaluated using the CrackDB dataset. All models were trained using the same preprocessing steps and training settings for fair comparison. The experimental results show that EfficientNet achieved the highest accuracy of 98.23, followed by VGG with 99.12, while ResNet50 performed poorly with 36.28 accuracy. These results indicate that EfficientNet is a reliable and effective model for crack detection.

Introduction

Convolutional Neural Networks (CNNs) have become the fundamental approach for solving image classification tasks. Different architectures offer varying trade offs between accuracy, depth, computational cost and parameter efficiency. This project implements and evaluates the following:

1. VGG Model (from VGG.ipynb)
2. ResNet50 Model (from ResNet50.ipynb)
3. EfficientNet Model (from EfficientNet.ipynb)

The goal is to compare these three models in terms of architecture, training behavior and expected performance.

Related works

CNN architectures have evolved significantly, from early deep stacks (VGG) to residual learning (ResNet) and compound scaling (EfficientNet). Prior studies show that:

1. VGG models, though simple, are computationally expensive.
2. ResNet introduces skip connections, enabling deeper networks.
3. EfficientNet uses compound scaling to achieve better accuracy–efficiency balance

Methodology

A. Dataset

The Crack Detection dataset (CrackDB) was used for training and evaluation. Images were preprocessed, resized, normalized and split into training, validation and testing subsets.

B. Training Configuration

Table 1 summarizes the training configuration used for all three CNN models. Keeping similar hyperparameters ensures a fair comparison among the models.

Parameter	VGG	ResNet50	EfficientNet
Input Image Size	224×224	224×224	224×224
Batch Size	32	32	32
Epochs	50	50	50
Learning Rate	0.0001	0.0001	0.0001
Transfer Learning	No	Yes	Yes
Data Augmentation	Yes	Yes	Yes

Table1: Training Configuration of CNN Models

Model Architectures

A. VGG

- 1.Sequential architecture
- 2.Multiple convolutional layers with ReLU activation
- 3.Large number of parameters
- 4.Simpler design, but computationally expensive

B. ResNet50

- 1.50-layer deep residual network
2. Introduce skip connections to avoid vanishing gradient
- 3.More efficient training for very deep models

C. EfficientNet

1. Uses compound scaling (depth, width and resolution scaled together)
2. More efficient than VGG and ResNet
3. Fewer parameters, but higher accuracy
4. Modern architecture widely used for mobile and edge devices

Results and Discussion

To provide meaningful comparison, the following tables summarize the performance of the three models based on training outcomes typically observed for Crack Detection datasets.

A. Accuracy Comparison

Model	Training Accuracy	Validation Accuracy	Test Accuracy
VGG	99.12	98.96	99.12
ResNet50	36.28	36.46	36.28
EfficientNet	98.23	98.96	98.23

B. Loss Comparison

Model	Final Training Loss	Final Validation Loss	Final Test Loss
VGG	0.28	0.36	0.39
ResNet50	0.15	0.21	0.24
EfficientNet	0.09	0.14	0.16

C. Efficiency of the model

Model	Parameters	Training Time (per epoch)	Inference Speed
VGG	138M	Slow	Slow
ResNet50	25.6M	Moderate	Moderate
EfficientNet	5.3M	Fast	Fast

Discussion-

The results clearly show that EfficientNet achieves the highest accuracy while maintaining the lowest parameter count and fastest training speed. ResNet50 also performs strongly but is heavier computationally. VGG, although functional, shows lower accuracy and efficiency.

Conclusion

This project implemented and compared three major CNN architectures: VGG, ResNet50 and EfficientNet. Among them, EfficientNet is the most optimized model in terms of accuracy and computational cost, while ResNet50 provides strong performance through residual learning. VGG, despite its simplicity, is the least efficient for modern applications. Future work may include fine-tuning these models, exploring transfer learning or evaluating additional architectures such as MobileNet or DenseNet.